Long-Term Survival Outcomes of Modified-FOLFOXIRI Plus Cetuximab Versus Bevacizumab in *RAS/BRAF* Wild-Type Metastatic Colorectal Cancer: Final Analysis of the DEEPER Trial

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ABSTRACT

The randomized phase II DEEPER trial (jRCTs061180022) previously reported superior depth of response with modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan (m-FOLFOXIRI) plus cetuximab versus bevacizumab in patients with *RAS* wild-type metastatic colorectal cancer (mCRC). Here, we report updated final survival outcomes and exploratory subgroup analyses focusing on *RAS/BRAF* wild-type and left-sided tumors. Final overall survival (OS) and progression-free survival (PFS) were analyzed in the per-protocol set (PPS) with exploratory subgroup analyses according to the presence of liver-limited disease and sex, using extended follow-up data. There were no differences in PFS and OS in the PPS. In patients with *RAS/BRAF* wild-type and left-sided tumors, median PFS was longer with cetuximab than with bevacizumab (14.8 *v* 11.9 months; hazard ratio [HR], 0.71 [95% CI, 0.52 to 0.97]; $P = .029$). The median OS was 50.2 months for cetuximab and 40.2 months for bevacizumab (HR, 0.74 [95% CI, 0.53 to 1.05]). In the exploratory analyses, cetuximab-based therapy was associated with longer OS in male patients (HR, 0.59; $P = .016$) and in patients with extrahepatic disease (HR, 0.60; $P = .014$). In conclusion, the DEEPER trial suggests that m-FOLFOXIRI plus cetuximab achieves superior tumor shrinkage and may be associated with favorable outcomes in selected patients with *RAS/BRAF* wild-type and left-sided mCRC, with exploratory signals of benefit in male patients and those with extrahepatic disease.

ACCOMPANYING CONTENT

Appendix

Data Sharing Statement

Protocol

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INTRODUCTION

Doublet combination chemotherapy with anti-epidermal growth factor receptor (EGFR) monoclonal antibodies is a standard approach for *RAS* wild-type and left-sided metastatic colorectal cancer (mCRC).[1-4] However, the clinical value of intensifying chemotherapy to a triplet backbone in combination with anti-EGFR therapy remains controversial. The DEEPER (JACCRO CC-13) trial investigated the efficacy of modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan (m-FOLFOXIRI) plus cetuximab compared with m-FOLFOXIRI plus bevacizumab in the first-line treatment of patients with *RAS* wild-type mCRC. In the primary analysis, cetuximab-based therapy was associated with a statistically significant improvement in depth of response (DpR). In patients with *RAS/BRAF* wild-type, left-sided primary mCRC, progression-free survival (PFS) and overall survival (OS) were significantly longer in the cetuximab group; however, because survival was longer than initially anticipated, the duration of follow-up was insufficient.[5] Therefore, the observation period was extended to 5 years to clarify the long-term clinical implications of intensive anti-EGFR–based triplet therapy. Here, we report the final survival results of the DEEPER trial with a particular focus on the subgroup with *RAS/BRAF* wild-type and left-sided tumors.

METHODS

DEEPER (UMIN000018217, jRCTs061180022) was a multicenter, randomized, open-label, phase II trial. Details on eligibility criteria and treatment schedules have been previously published.[5] Full inclusion and exclusion criteria are available in the study protocol.

Written and informed consent was obtained from all participants before initiating trial screening procedures and enrollment. The trial was approved by a certified review board and conducted in compliance with the protocol and the Japanese Clinical Trials Act (Act No. 16 of 14 April 2017).

The statistical design and primary analysis of the DEEPER trial have been reported previously.[5] Briefly, the study was designed to evaluate the superiority of m-FOLFOXIRI plus cetuximab over bevacizumab-based therapy in terms of DpR in the per-protocol set (PPS), defined according to prespecified criteria for evaluability of DpR. This report focuses on the final analysis of PFS and OS with extended follow-up. Exploratory subgroup analyses were conducted according to clinical factors including tumor sidedness, *BRAF* V600E status, sex, and metastatic pattern, but not adjusted for multiplicity. Data on *BRAF* mutation status were retrospectively collected from clinical records and available molecular testing.

The efficacy was evaluated in the PPS, and survival outcomes were estimated using the Kaplan-Meier method and compared with the log-rank test. Hazard ratios (HRs) and corresponding 95% CIs were estimated using Cox proportional hazards models. All statistical tests were two-sided, and *P* values ≤.05 were considered statistically significant.

During the conduct of the study, the protocol was amended to extend the planned follow-up period from 3 to 5 years after patient enrollment to allow for adequate assessment of long-term survival outcomes, given the longer-than-anticipated survival observed in the study population. The extended data cutoff for the present analysis was August 2024.

RESULTS

Between July 2015 and June 2019, a total of 359 patients with *RAS* wild-type mCRC were enrolled. Finally, 86 and 92 patients were evaluable for updated analysis of *RAS*/*BRAF* wild-type and left-sided tumors in the cetuximab arm and the bevacizumab arm, respectively (Fig 1).

Baseline demographic and disease characteristics were well balanced between treatment arms in PPS and subgroup population with *RAS*/*BRAF* wild-type and left-sided tumors (Table 1 and Appendix Table A1, online only). About 40% patients received surgery as post-treatment after protocol therapy in both treatment arms (Appendix Table A2). At the time of the final analysis, the median duration of follow-up was 75.8 months.

In the overall PPS population with *RAS* wild-type tumors, no statistically significant differences in survival outcomes were observed between the two treatment arms. When stratified by primary tumor location, distinct survival patterns emerged (Appendix Table A3, and Figs A1 and A2).

In the exploratory subgroup of patients with *RAS*/*BRAF* wild-type and left-sided tumors (n = 178), m-FOLFOXIRI plus cetuximab was associated with a statistically significant improvement in PFS compared with bevacizumab-based therapy. The median PFS was 14.8 months in the cetuximab arm versus 11.9 months in the bevacizumab arm (HR,

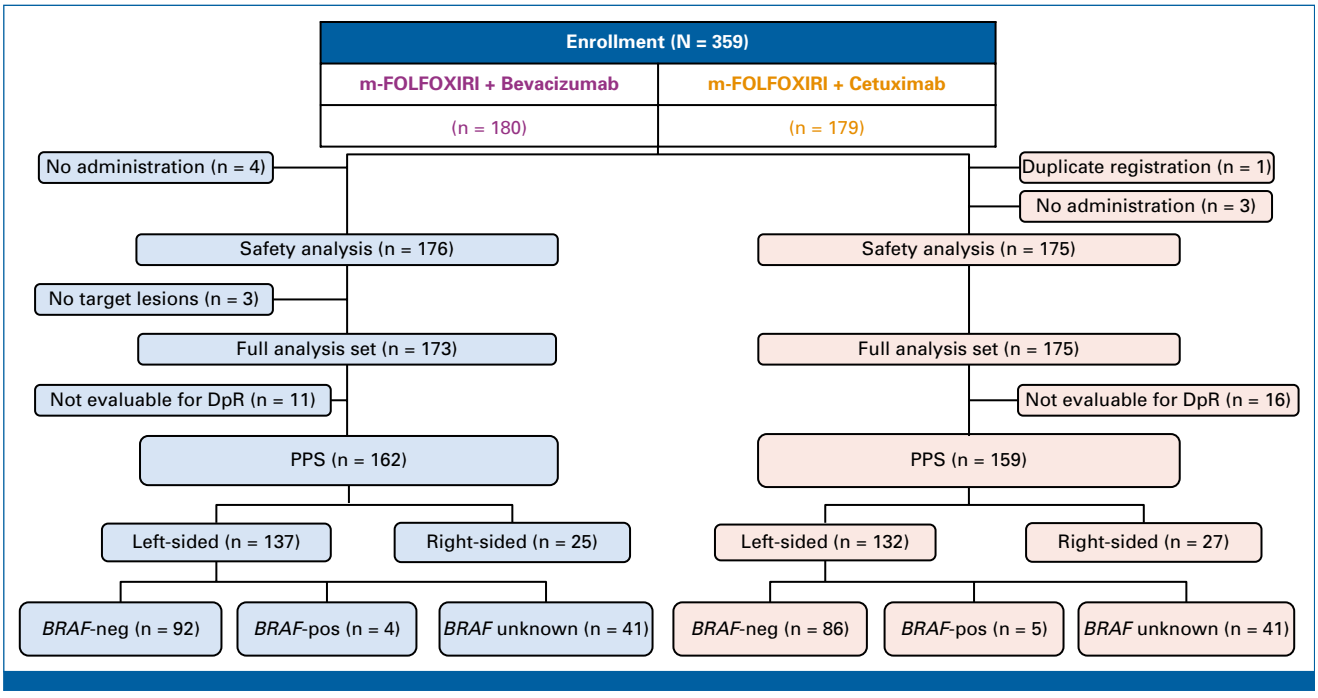


FIG 1. Trial profile. DpR, depth of response; m-FOLFOXIRI, modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan; PPS, per-protocol set.

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TABLE 1. Baseline Characteristics in Subgroup Population With *RAS*/*BRAF* Wild-Type and Left-Sided Tumors

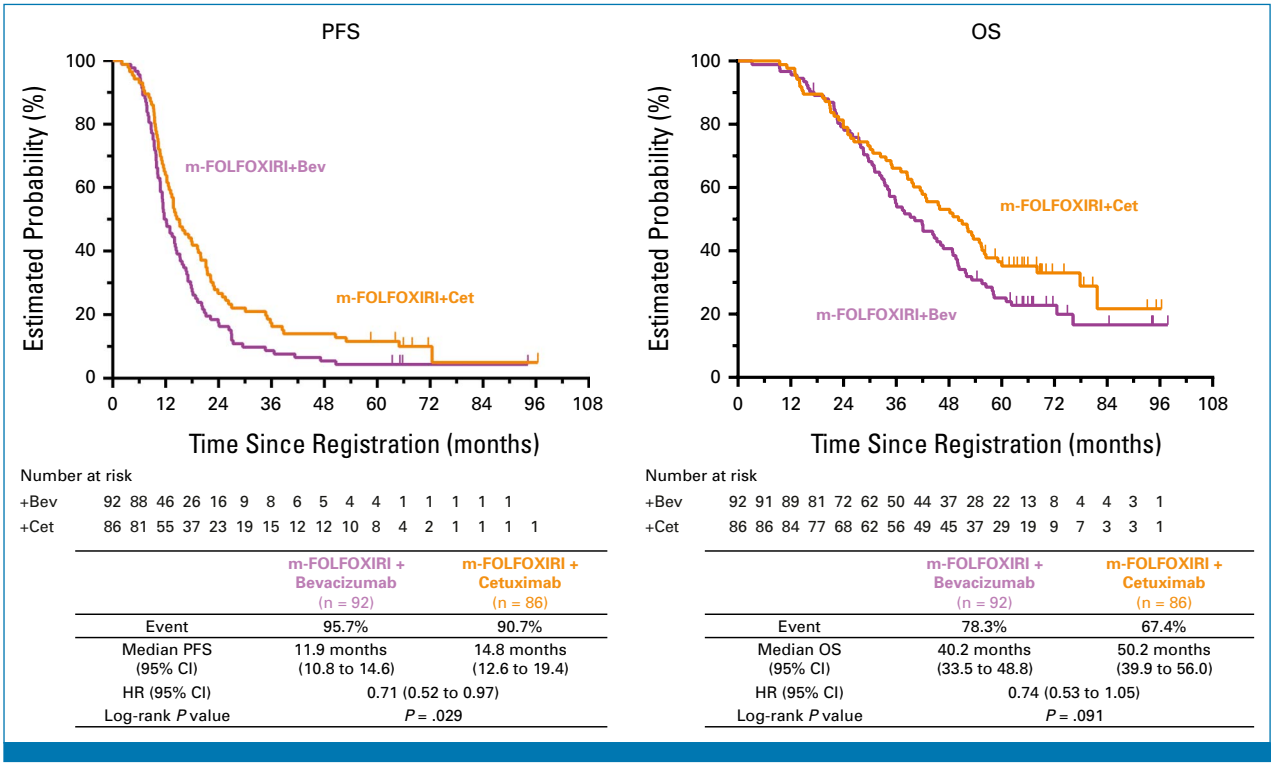
Characteristic	m-FOLFOXIRI + Cetuximab (n = 86)	m-FOLFOXIRI + Bevacizumab (n = 92)
Sex, No. (%)		
Male/female	62 (72.1)/24 (27.9)	57 (62.0)/35 (38.0)
Age, years, median (range)	64.5 (26-83)	64.5 (29-78)
Adjuvant chemotherapy, No. (%)		
None/with oxaliplatin/without oxaliplatin	83 (96.5)/1 (1.2)/2 (2.3)	89 (96.7)/2 (2.2)/1 (1.1)
ECOG performance status, No. (%)		
0/1	79 (91.9)/7 (8.1)	85 (92.4)/7 (7.6)
Diagnosis, No. (%)		
Advanced/relapsed	77 (89.5)/9 (10.5)	77 (83.7)/15 (16.3)
Primary lesion, No. (%)		
None/present	43 (50.0)/43 (50.0)	50 (54.3)/42 (45.7)
No. of metastatic sites, No. (%)		
1-1 ≥2	30 (34.9)/56 (65.1)	44 (47.8)/48 (52.2)
Liver-only disease, No. (%)		
Yes/No	21 (24.4)/65 (75.6)	32 (34.8)/60 (65.2)
UGT1A1 (*6, *28), No. (%)		
Wild/hetero/homo and double hetero	35 (40.7)/29 (33.7)/4 (4.7)	41 (44.6)/26 (28.3)/12 (13.0)

Abbreviations: ECOG, Eastern Cooperative Oncology Group; m-FOLFOXIRI, modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan.

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0.71 [95% CI, 0.52 to 0.97]; $P = .029$). The median OS in this subgroup was 50.2 months with cetuximab and 40.2 months with bevacizumab (HR, 0.74 [95% CI, 0.53 to 1.05]; $P = .091$; Fig 2 and Table 2). Conversely, in patients with *RAS*/*BRAF* wild-type but right-sided tumors, no survival benefit was observed with cetuximab-based therapy (Appendix Table A3).

Exploratory subgroup analyses within the *RAS*/*BRAF* wild-type and left-sided population suggested heterogeneity of



	m-FOLFOXIRI + Bevacizumab (n = 92)	m-FOLFOXIRI + Cetuximab (n = 86)
Event	95.7%	90.7%
Median PFS (95% CI)	11.9 months (10.8 to 14.6)	14.8 months (12.6 to 19.4)
HR (95% CI)	0.71 (0.52 to 0.97)	
Log-rank *P* value	*P* = .029	

	m-FOLFOXIRI + Bevacizumab (n = 92)	m-FOLFOXIRI + Cetuximab (n = 86)
Event	78.3%	67.4%
Median OS (95% CI)	40.2 months (33.5 to 48.8)	50.2 months (39.9 to 56.0)
HR (95% CI)	0.74 (0.53 to 1.05)	
Log-rank *P* value	*P* = .091	

FIG 2. Progression-free survival and OS in patients with *RAS*/*BRAF* wild-type, left-sided, metastatic colorectal cancer. HR, hazard ratio; m-FOLFOXIRI, modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan; OS, overall survival; PFS, progression-free survival.

TABLE 2. Treatment Outcomes in Patients With *RAS*/*BRAF* Wide-Type and Left-Sided Tumors

End Point	m-FOLFOXIRI + Cetuximab (n = 86)	m-FOLFOXIRI + Bevacizumab (n = 92)
Median DpR, % (range)	59.2 (3.6-100)	47.5 (6.9-100)
Mean, % (95% CI)	59.6 (54.7 to 64.5)	49.4 (45.4 to 53.4)
Standard deviation (95% CI)	22.9 (19.9 to 26.9	19.3 (16.9 to 22.6)
Welch *t*-test	*P* = .0017	
Objective response rate, % (95% CI)	75.6 (66.5 to 84.7)	72.8 (63.7 to 81.9)
Chi-squared test	*P* = .67	
Disease control rate, % (95% CI)	100 (100 to 100)	100 (100 to 100)
Chi-squared test	*P* = NA	
R0 resection rate, % (95% CI)	31.4 (21.6 to 41.2)	22.8 (14.2 to 31.4)
Chi-squared test	*P* = .20	
Median progression-free survival, months	14.8	11.9
HR (95% CI), Log-rank *P*-value	0.71 (0.52 to 0.97), *P* = .029	
Median overall survival, months	50.2	40.2
HR (95% CI), Log-rank *P*-value	0.74 (0.53 to 1.05), *P* = .091	

Abbreviations: DpR, depth of response; HR, hazard ratio; m-FOLFOXIRI, modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan; NA, not available.

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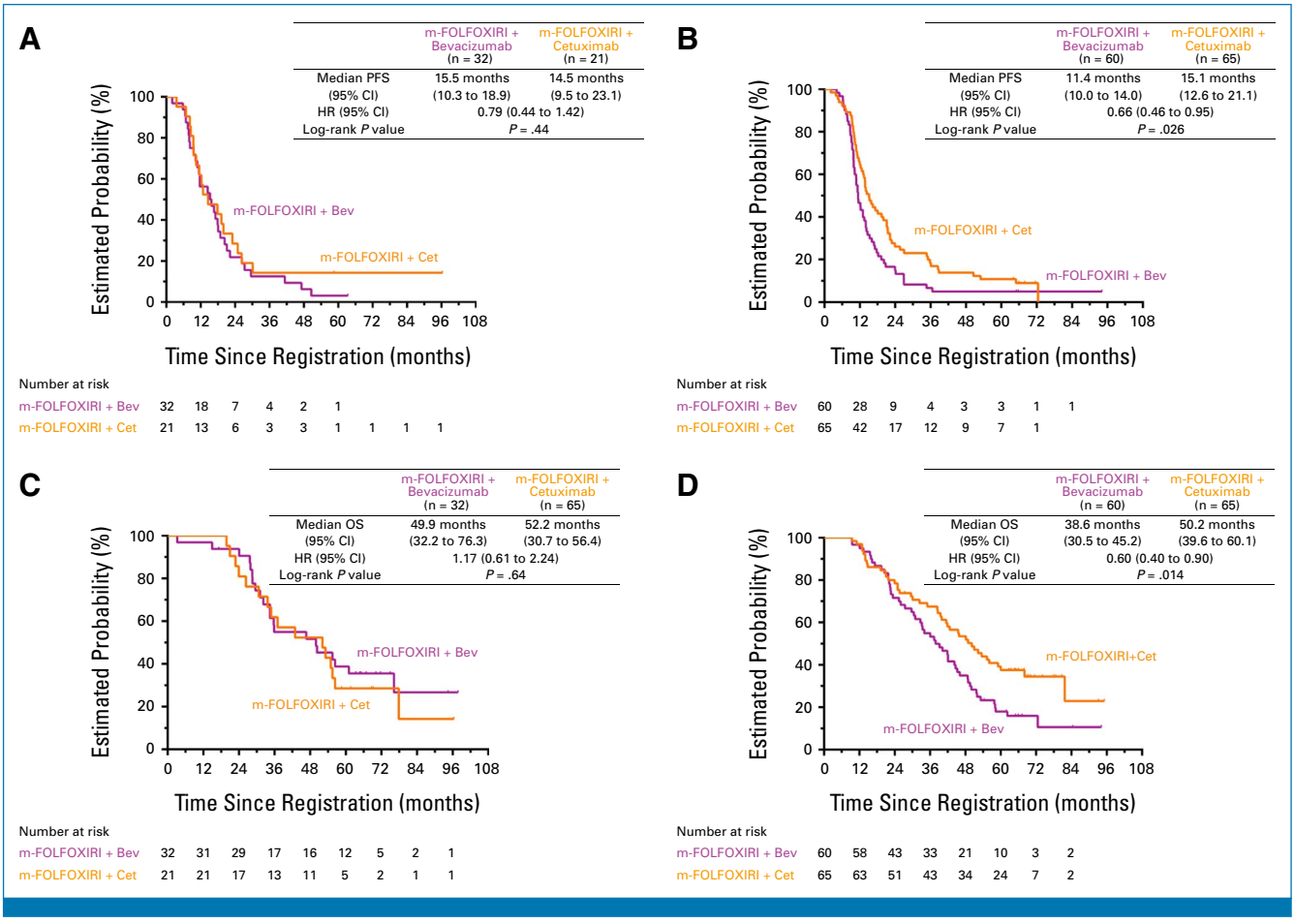


FIG 3. PFS and OS by liver disease status in patients with *RAS*/*BRAF* wild-type, left-sided, metastatic colorectal cancer; (A) PFS in patients with liver-limited disease; (B) PFS in patients with extrahepatic metastatic disease; (C) OS in patients with liver-limited disease; (D) OS in patients with extrahepatic metastatic disease. HR, hazard ratio; m-FOLFOXIRI, modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan; OS, overall survival; PFS, progression-free survival.

treatment effect according to clinical factors (Appendix Fig A3). Cetuximab-based therapy was associated with a trend toward longer OS in patients with extrahepatic metastatic disease (interaction $P = .08$; Fig 3). In addition, a potential interaction was observed between treatment effect and sex. In male patients, the median OS was 52.8 months in the cetuximab arm and 40.2 months in the bevacizumab arm (HR, 0.59 [95% CI, 0.38 to 0.91]), whereas no survival advantage was observed in female patients (interaction $P = .05$) (Appendix Fig A4).

The incidence of grade 3 to 4 adverse events was comparable between treatment arms. No new or unexpected safety signals were identified in the final analysis (Appendix Table A4).

DISCUSSION

The final survival analysis of the DEEPER trial provides a comprehensive assessment of the long-term clinical impact of intensifying first-line therapy with m-FOLFOXIRI plus cetuximab in patients with *RAS*/*BRAF* wild-type and left-sided mCRC. With extended follow-up exceeding 5 years, several clinically important insights emerge that refine the interpretation of the primary analysis and inform patient selection for intensive anti-EGFR–based strategies.

The survival outcomes observed with bevacizumab-based triplet therapy in DEEPER were comparable with those reported in molecular subgroup analyses of the TRIBE study,[6] suggesting that the enrolled population was representative and that the observed benefit with cetuximab in selected subgroups is unlikely to be attributable solely to favorable baseline characteristics. The updated analysis of the TRIPLETE trial showed significantly improved OS in the triplet group,[7] supporting the findings of the DEEPER trial.

Exploratory analyses suggest heterogeneity of treatment effect according to clinical characteristics, including metastatic pattern and sex. In mCRC with liver-limited metastases, previous clinical trials have demonstrated no difference between anti-EGFR antibody drugs and bevacizumab-based chemotherapy.[8,9] For patients with non–liver-limited metastases, such as those with systemic metastases, implementing regimens with strong tumor-shrinking effects has been shown to prolong survival. Previous studies have suggested greater efficacy of anti-EGFR therapy in male patients.[10,11] In our trial, female patients experienced a higher incidence of skin toxicity (94% *v* 75%) and tended to receive lower dose intensity. Given that cetuximab-mediated antibody-dependent cellular cytotoxicity is partly mediated by natural killer (NK) cells, and that NK cell counts are higher in male patients[12] and decline with age,[13] these factors may contribute to the observed sex-based differences.

Several limitations of this study should be acknowledged. First, the DEEPER trial was a phase II study and was not statistically powered to detect differences in OS; accordingly, all subgroup analyses were exploratory and were conducted without adjustment for multiplicity. Second, *BRAF* mutational status was not available for all patients at the time of enrollment, reflecting real-world clinical practice at the initiation of the trial. Third, subsequent lines of therapy, including crossover to alternative biologic agents, may have confounded the interpretation of long-term survival outcomes. Nevertheless, the maturity of follow-up and the consistency of results across related clinical end points support the robustness of the overall conclusions.

In conclusion, the final analysis of the DEEPER trial suggests that m-FOLFOXIRI plus cetuximab achieves superior tumor shrinkage and may be associated with favorable long-term outcomes in selected patients with *RAS*/*BRAF* wild-type and left-sided mCRC, whereas no clear survival advantage was observed in the overall *RAS* wild-type population.

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CLINICAL TRIAL INFORMATION

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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DATA SHARING STATEMENT

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Manuscript writing: All authors
Final approval of manuscript: All authors
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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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APPENDIX

TABLE A1. Baseline Characteristics in Per-Protocol Population With *RAS* Wild-Type Tumors

Characteristic	m-FOLFOXIRI + Bevacizumab (n = 162)	m-FOLFOXIRI + Cetuximab (n = 159)
Sex, No. (%)		
Male/female	106 (65.4)/56 (34.6)	101 (63.5)/58 (36.5)
Age, years		
Median (range)	65 (29-85)	65 (26-83)
Site of primary tumor, No. (%)		
Right/left	25 (15.4)/137 (84.6)	27 (17.0)/132 (83.0)
Adjuvant chemotherapy, No. (%)		
None/with oxaliplatin/without oxaliplatin	152 (93.8)/6 (3.7)/4 (2.5)	148 (93.1)/7 (4.4)/4 (2.5)
ECOG performance status, No. (%)		
0/1	146 (90.1)/16 (9.9)	146 (91.8)/13 (8.2)
Diagnosis, No. (%)		
Advanced/relapsed	136 (84.0)/26 (16.0)	136 (85.5)/23 (14.5)
Primary lesion, No. (%)		
−/+	86 (53.1)/76 (46.9)	83 (52.2)/76 (47.8)
Liver-only disease, No. (%)		
Yes/no	50 (30.9)/112 (69.1)	39 (24.5)/120 (75.5)
UGT1A1 (*6, *28), No. (%)		
Wild/hetero/homo and double hetero	69 (42.6)/46 (28.4)/22 (13.6)	65 (40.9)/59 (37.1)/9 (5.7)

Abbreviation: ECOG, Eastern Cooperative Oncology Group.

TABLE A2. Dose Intensity and Post-Treatment in Each Treatment Arm

Treatment	m-FOLFOXIRI + Bevacizumab (n = 92)	m-FOLFOXIRI + Cetuximab (n = 86)
Treatment courses, median (range)	12 (3-68)	13 (2-87)
Dose intensity, % (induction phase)		
Bevacizumab	71.2	—
Cetuximab	—	78.2
Irinotecan	66.5	67.2
Oxaliplatin	62.3	63.7
5-FU	76.8	75.5
Dose intensity, % (maintenance phase)		
Bevacizumab	79.3	—
Cetuximab	—	70.5
5-FU	86.2	91.1
Post-treatment, No. (%)		
Chemotherapy/surgery/others	50 (54.3)/38 (41.3)/1 (1.1)	41 (47.7)/39 (45.3)/1 (1.2)

Abbreviations: 5-FU, 5-fluorouracil; m-FOLFOXIRI, modified 5-FU, leucovorin, oxaliplatin and irinotecan.

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TABLE A3. Clinical Outcomes of m-FOLFOXIRI + Cetuximab Versus Bevacizumab According to *RAS/BRAF* Status and Primary Tumor Location

			Left Side		Right Side	
End Point	m-FOLFOXIRI + Bevacizumab	m-FOLFOXIRI + Cetuximab	m-FOLFOXIRI + Bevacizumab	m-FOLFOXIRI + Cetuximab	m-FOLFOXIRI + Bevacizumab	m-FOLFOXIRI + Cetuximab
RAS wild-type, n	162	159	137	132	25	27
Median DpR, % (range)	46.0 (−0.6 to 100)	57.3 (−42.6 to 100)	46.1 (3.2 to 100)	59.2 (−42.6 to 100)	41.2 (−0.6 to 85.6)	50.0 (−15.0 to 100)
Welch *t*-test	$P = .0029$		$P = .0026$		$P = .47$	
Objective response rate, % (95% CI)	69.1 (62.0 to 76.2)	71.1 (64.0 to 78.1)	70.8 (63.2 to 78.4)	74.2 (66.8 to 81.7)	60.0 (40.8 to 79.2)	55.6 (36.8 to 74.3)
Chi-squared test	$P = .71$		$P = .53$		$P = .75$	
Disease control rate, % (95% CI)	99.4 (98.2 to 100)	98.7 (97.0 to 100)	99.3 (97.8 to 100)	98.5 (96.4 to 100)	100.0 (100 to 100)	100.0 (100 to 100)
Chi-squared test	$P = .55$		$P = .54$		$P =$ NA	
Median PFS, months	12.3	13.0	12.1	13.9	12.8	9.0
HR (95% CI), *P*	0.88 (0.70 to 1.11), $P = .29$		0.81 (0.63 to 1.05), $P = 0.11$		1.46 (0.84 to 2.54), $P = 0.18$	
Median OS, months	41.6	41.7	41.9	45.3	33.9	28.4
HR (95% CI), *P*	0.92 (0.71 to 1.19), $P = .52$		0.85 (0.64 to 1.12), $P = .25$		1.37 (0.75 to 2.54), $P = .31$	
RAS/BRAF wild-type, n	107	105	92	86	15	19
Median DpR, % (range)	47.9 (−0.6 to 100)	58.0 (−15.0 to 100)	47.5 (6.9 to 100)	59.2 (3.6 to 100)	58.9 (−0.6 to 85.6)	53.3 (−15.0 to 100)
Welch *t*-test	$P = .012$		$P = .0017$		$P = .74$	
Objective response rate, % (95% CI)	72.9 (64.5 to 81.3)	72.4 (63.8 to 80.9)	72.8 (63.7 to 81.9)	75.6 (66.5 to 84.7)	73.3 (51.0 to 95.7)	57.9 (35.7 to 80.1)
Chi-squared test	$P = .93$		$P = .67$		$P = .35$	
Disease control rate, % (95% CI)	100.0 (100 to 100)	100.0 (100 to 100)	100.0 (100 to 100)	100.0 (100 to 100)	100.0 (100 to 100)	100.0 (100 to 100)
Chi-squared test	$P =$ NA		$P =$ NA		$P =$ NA	
Median PFS, months	13.1	13.8	11.9	14.8	14.7	9.7
HR (95% CI), *P*	0.83 (0.63 to 1.10), $P = .19$		0.71 (0.52 to 0.97), $P = .029$		2.13 (1.04 to 4.37), $P = .034$	
Median OS, months	42.1	45.8	40.2	50.2	47.3	31.6
HR (95% CI), *P*	0.85 (0.62 to 1.16), $P = .29$		0.74 (0.53 to 1.05), $P = .091$		1.62 (0.74 to 3.54), $P = .22$	

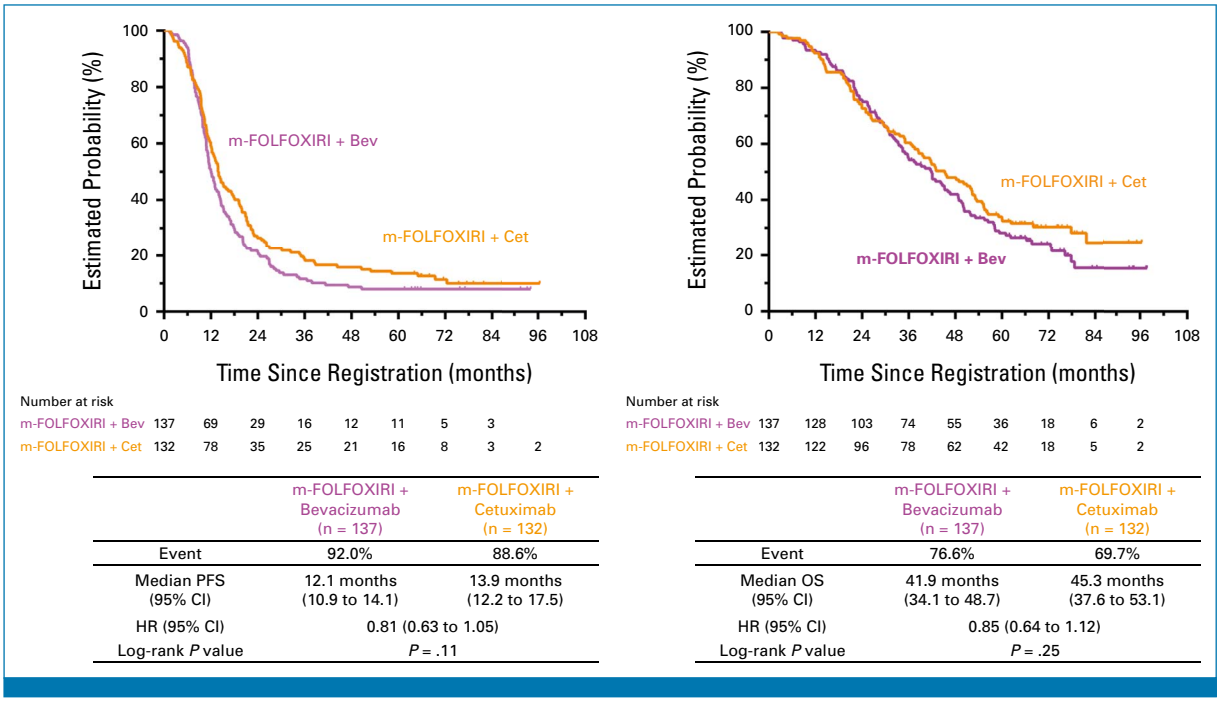
Abbreviations: DpR, depth of response; HR, hazard ratio; m-FOLFOXIRI, modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan; NA, not available; OS, overall survival; PFS, progression-free survival.

TABLE A4. Safety Profile in Each Treatment Arm

	m-FOLFOXIRI + Bevacizumab (n = 176)		m-FOLFOXIRI + Cetuximab (n = 175)	
Adverse Event	≥ Grade 1, No. (%)	≥ Grade 3, No. (%)	≥ Grade 1, No. (%)	≥ Grade 3, No. (%)
Neutropenia	154 (87.5)	96 (54.5)	154 (88.0)	98 (56.0)
Febrile neutropenia	19 (10.8)	19 (10.8)	15 (8.6)	15 (8.6)
Thrombocytopenia	71 (40.3)	2 (1.1)	82 (46.9)	4 (2.3)
Anemia	102 (58.0)	8 (4.5)	90 (51.4)	11 (6.3)
Mucositis oral	75 (42.6)	4 (2.3)	102 (58.3)	17 (9.7)
Anorexia	99 (56.3)	19 (10.8)	108 (61.7)	21 (12.0)
Nausea	89 (50.6)	11 (6.3)	83 (47.4)	11 (6.3)
Vomiting	31 (17.6)	3 (1.7)	18 (10.3)	1 (0.6)
Diarrhea	101 (57.4)	14 (8.0)	106 (60.6)	21 (12.0)
Fatigue	64 (36.4)	11 (6.3)	58 (33.1)	12 (6.9)
Rash acneiform	5 (2.8)	0 (0.0)	134 (76.6)	23 (13.1)
Paronychia	2 (1.1)	0 (0.0)	101 (57.7)	20 (11.4)
Pruritus	0 (0.0)	0 (0.0)	30 (17.1)	3 (1.7)
Alopecia	70 (39.8)	0 (0.0)	64 (36.6)	0 (0.0)
Peripheral sensory neuropathy	127 (72.2)	8 (4.5)	131 (74.9)	12 (6.9)
Hypomagnesemia	31 (17.6)	0 (0.0)	99 (56.6)	7 (4.0)
Proteinuria	92 (52.3)	6 (3.4)	80 (45.7)	3 (1.7)
Hypertension	124 (70.5)	59 (33.5)	85 (48.6)	31 (17.7)
Thromboembolic event	14 (8.0)	3 (1.7)	13 (7.4)	6 (3.4)

Abbreviation: m-FOLFOXIRI, modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan.

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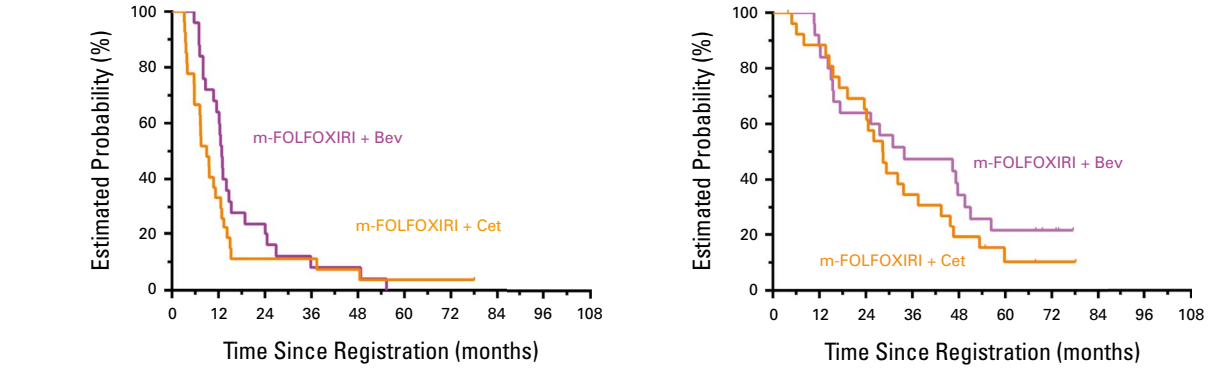


	m-FOLFOXIRI + Bevacizumab (n = 137)	m-FOLFOXIRI + Cetuximab (n = 132)
Event	92.0%	88.6%
Median PFS (95% CI)	12.1 months (10.9 to 14.1)	13.9 months (12.2 to 17.5)
HR (95% CI)	0.81 (0.63 to 1.05)	
Log-rank *P* value	*P* = .11	

	m-FOLFOXIRI + Bevacizumab (n = 137)	m-FOLFOXIRI + Cetuximab (n = 132)
Event	76.6%	69.7%
Median OS (95% CI)	41.9 months (34.1 to 48.7)	45.3 months (37.6 to 53.1)
HR (95% CI)	0.85 (0.64 to 1.12)	
Log-rank *P* value	*P* = .25	

FIG A1. PFS and OS in *RAS* wild-type, left-side. HR, hazard ratio; m-FOLFOXIRI, modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan; OS, overall survival; PFS, progression-free survival.

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	m-FOLFOXIRI + Bevacizumab (n = 25)	m-FOLFOXIRI + Cetuximab (n = 27)
Event	100%	96.3%
Median PFS (95% CI)	12.8 months (10.7 to 15.3)	9.0 months (5.8 to 12.6)
HR (95% CI)	1.46 (0.84 to 2.54)	
Log-rank *P* value	*P* = .18	

	m-FOLFOXIRI + Bevacizumab (n = 25)	m-FOLFOXIRI + Cetuximab (n = 27)
Event	76.0%	85.2%
Median OS (95% CI)	33.9 months (15.7 to 49.6)	28.4 months (19.3 to 37.5)
HR (95% CI)	1.37 (0.75 to 2.54)	
Log-rank *P* value	*P* = .31	

FIG A2. PFS and OS in RAS wild-type, right-side. HR, hazard ratio; m-FOLFOXIRI, modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan; OS, overall survival; PFS, progression-free survival.

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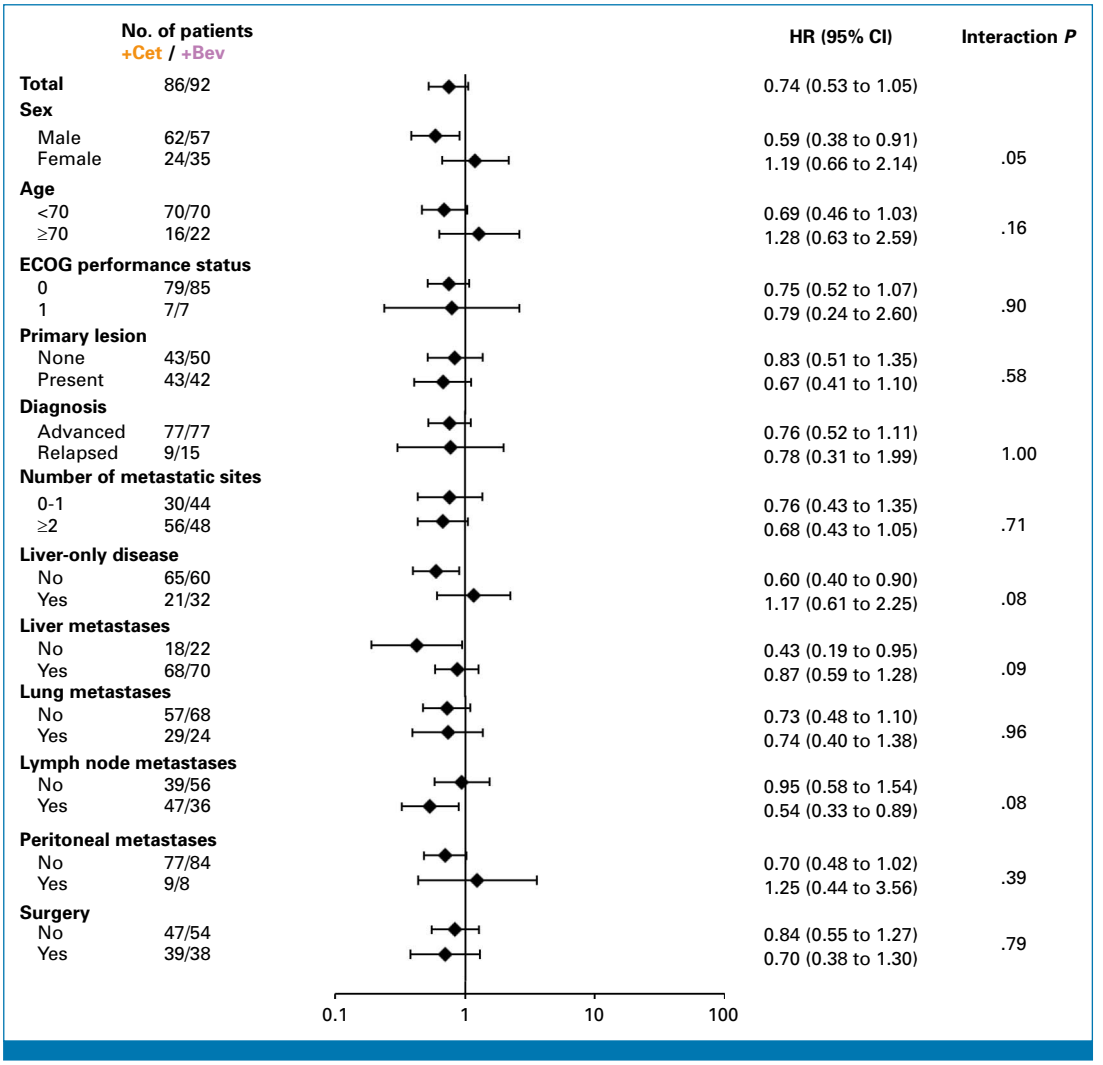
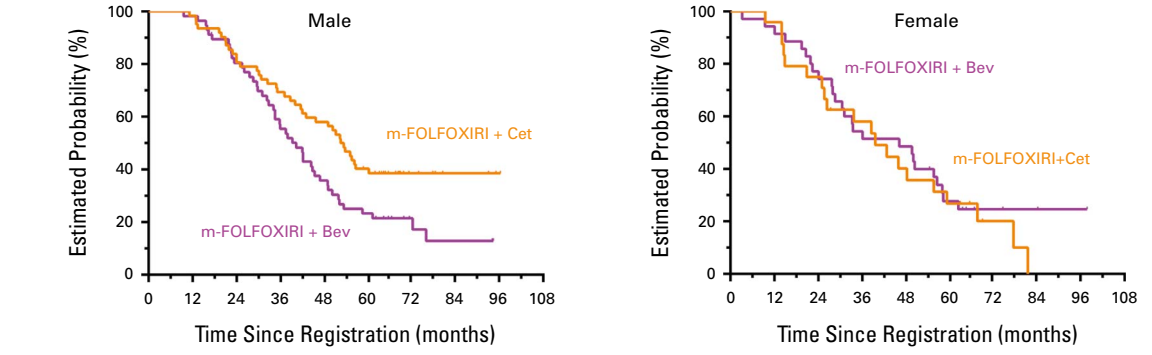


FIG A3. OS forest plot. ECOG, Eastern Cooperative Oncology Group; HR, hazard ratio; OS, overall survival.

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Number at risk

m-FOLFOXIRI + Bev	57	56	45	31	20	13	5	2	
m-FOLFOXIRI + Cet	62	61	50	43	36	23	7	3	1

Number at risk

m-FOLFOXIRI + Bev	35	33	27	19	17	9	3	2	1
m-FOLFOXIRI + Cet	24	23	18	13	9	6	2		

	m-FOLFOXIRI + Bevacizumab (n = 57)	m-FOLFOXIRI + Cetuximab (n = 62)
Event	80.7%	61.3%
Median OS (95% CI)	40.2 months (34.1 to 45.2)	52.8 months (41.5 to NR)
HR (95% CI)	0.59 (0.38 to 0.91)	
Log-rank *P* value	*P* = .016	

Interaction *P* value .05

	m-FOLFOXIRI + Bevacizumab (n = 35)	m-FOLFOXIRI + Cetuximab (n = 24)
Event	74.3%	83.3%
Median OS (95% CI)	46.1 months (28.6 to 56.4)	39.6 months (25.0 to 59.1)
HR (95% CI)	1.19 (0.67 to 2.14)	
Log-rank *P* value	*P* = .55	

FIG A4. Overall survival by sex in RAS/BRAF wild-type, left-side. HR, hazard ratio; m-FOLFOXIRI, modified 5-fluorouracil, leucovorin, oxaliplatin and irinotecan; NR, not available; OS, overall survival.

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